

Xiang Dai

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SUMMARY

Ph.D. candidate in Electrical & Computer Engineering with expertise in optical design, computational imaging, imaging processing, and hardware–software integration. Experienced in sensor characterization and optimization algorithms.

EDUCATION

UC San Diego

Ph.D., Electrical and Computer Engineering

La Jolla, CA

Sep 2021 – Spring 2027

Duke University

Master of Science, Biomedical Engineering (GPA: 3.8)

Durham, NC

Aug 2019 - Jun 2021

Changchun University of Sci. & Tech. (CUST)

Bachelor of Engineering, Electro-Optical Engineering (GPA: 3.95/5, Top 5/205)

Changchun, China

Sep 2015 - Jun 2019

SKILLS

- **Programming & Data Analysis:** Python, MATLAB
- **Optical Systems & Prototyping:** Zemax, OpenSCAD, optical system assembly, sensor characterization
- **Automation & Optimization:** Hardware–software integration, convex & non-convex optimization, image processing

RESEARCH EXPERIENCE

Particle Vertex Reconstruction in 3D

Advisor: Nick Antipa, Felicia Sutanto, Jingke Xu

UC San Diego, Lawrence Livermore National Lab

May 2024 – March 2026

- Prototyped a multi-focal lens array enabling 3D particle vertex localization within a 5 cm³ volume.
- Developed rendering-based forward models and Fisher Information analysis to establish theoretical system sensitivity.
- Applied the ADCG algorithm for 3D particle reconstruction.

Single-shot HDR using conventional image sensor shutter functions and optical randomization

Advisor: Nick Antipa

UC San Diego

Sep 2021 - May 2024

- Designed and prototyped a single-shot HDR system using a customized random fiber bundle and off-the-shelf optics.
- Developed FISTA with total variation regularization, achieving a dynamic range of up to 73 dB from an 8-bit input.

High-speed video from rolling & global shutter measurements

Advisor: Nick Antipa

UC San Diego

Sep 2024 - March 2025

- Proposed a dual-sensor hardware design combining a diffuser camera with a conventional camera for high-speed video.
- Prototyped the dual-sensor, achieving 4,800 frames per second at 10x lower cost than existing systems.

Quantitative Jones matrix imaging using vectorial Fourier ptychography

Advisor: Roarke Horstmeyer

Duke University

Aug 2019 - June 2021

- Extended scalar Fourier ptychography to a vectorial model based on Jones calculus to capture full polarization effects.
- Demonstrated the method experimentally, achieving ~29 mm² field-of-view reconstructions at 1.24 μm resolution, quantitatively mapping retardance, diattenuation, and orientation.

WORKING EXPERIENCE

Advanced Development Internship

Manager: Matthew Andrew

Zeiss Research Microscopy Solutions

March 2026 – Present

- Phase contrast X-ray microscopy with 2D grating system development.

PATENT

Dai, X., et al. "Method and system of polarization microscopy imaging." U.S. Patent Application No. 18/073,759.

SELECTED PUBLICATIONS

Dai, X., et al. (2025). Single-shot HDR using conventional image sensor shutter functions and optical randomization. ACM TOG (Invited to present at Siggraph)

Dai, X.*, et al. (2025) "RnGCam: High-speed video from rolling & global shutter measurements." ICCV. (*co-first authors)

Dai, X., et al. (2022). Quantitative Jones matrix imaging using vectorial Fourier ptychography. Biomedical optics express, 13(3), 1457-1470. (Editor's pick)